Full Molecular Hamiltonian

From Classical to Quantum: the exact non-relativistic quantization

Chih-En Shen

Roadmap: four stages

We build the exact non-relativistic molecular Hamiltonian in four stages:

1. **Classical** — lab-frame H; separate the centre of mass.
2. **Body-Fixed** — co-rotating frame + Routhian $\Rightarrow$ classical molecular H.
3. **Quantize** — Podolsky rule, anomalous algebra, Watson pseudopotential.
4. **Quantum** $\hat{H}$ — the exact non-relativistic molecular Hamiltonian.

Outline

Motivation

Why is the molecular Hamiltonian non-trivial?

The lab Hamiltonian couples *all* particles through the Coulomb potential:

$$H = \sum_j \frac{\mathbf{P}_j^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V.$$

To do molecular physics we want to separate

$$\underbrace{\text{translation}}_{\text{COM}} \;\Big|\; \underbrace{\text{rotation}}_{\text{Euler angles }\Omega} \;\Big|\; \underbrace{\text{vibration}}_{\text{normal coords }Q} \;\Big|\; \underbrace{\text{electrons}}_{\bar{\mathbf{r}}_i}.$$

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The catch

Rotation + vibration are *curvilinear* coordinates. Naively quantizing $p \to -i\hbar\,\partial$ is **not** Hermitian there. Getting it right forces the Podolsky construction, an *anomalous* angular-momentum algebra, and the Watson pseudopotential.

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We keep everything exact until the very end:

- **No** Born–Oppenheimer / clamped nuclei.
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The *only* ingredients are:

1. non-relativistic dynamics;
2. the Eckart frame (a *convention*, not an approximation);
3. parametrizing the nuclear shape by $3N_{\mathrm{nucl}} - 6$ internal coordinates (exact).

The endpoint is the exact quantum $\hat{H}_{\mathrm{int}}$ — the rigorous starting point for any later approximation.

Classical Cartesian Hamiltonian

The starting point (lab frame)

N_{nucl} nuclei $(M_j, Z_j e, \mathbf{R}_j, \mathbf{P}_j)$ and N_{elec} electrons $(m_e, -e, \mathbf{r}_i, \mathbf{p}_i)$, all in the space-fixed frame:

$$H = \underbrace{\sum_j \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_i \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \tag{1.1}$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{i\neq j}\frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j}\frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}}\right). \tag{1.2}$$

Canonical brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk}\delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij}\delta_{\alpha\beta}$.

Everything that follows is an exact rewriting of (1.1)–(1.2).

Nuclear centre-of-mass separation

New coordinates: nuclear COM + relative

Split off the nuclear centre of mass and refer electrons to it:

$$\mathbf{R}_{\mathrm{cm}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j, \qquad M_N = \sum_j M_j, \tag{2.1}$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\mathrm{cm}}, \qquad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\mathrm{cm}}. \tag{2.2}$$

The relative nuclear vectors are linearly dependent:

$$\sum_j M_j \mathbf{R}'_j = 0. \tag{2.3}$$

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A point (F_2-)transformation gives the conjugate momenta [dual constraint $\sum_k \mathbf{P}'_k = 0$]:

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_N}\left(\mathbf{P}_{\mathrm{cm}} - \textstyle\sum_i \mathbf{p}_i\right) + \mathbf{P}'_j, \qquad \mathbf{p}_i = \mathbf{p}'_i} \tag{2.7}$$

with $\mathbf{P}_{\mathrm{cm}} = \sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i$ the *total* momentum.

Hamiltonian after the transformation

Inserting (2.7) and expanding the nuclear kinetic energy,

$$H = \frac{\mathbf{P}_{\mathrm{cm}}^2}{2M_N} \underbrace{- \frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \mathbf{p}_i}{M_N}}_{\text{artifact}} + \underbrace{\frac{(\sum_i \mathbf{p}_i)^2}{2M_N}}_{\text{mass polarization}} + \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V. \tag{2.10}$$

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Setting $\mathbf{P}_{\mathrm{cm}} = 0$ (rest frame) gives the internal Hamiltonian

$$\boxed{H_{\mathrm{int}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + \underbrace{\frac{1}{2M_N}\Big(\sum_i \mathbf{p}_i\Big)^2}_{\text{electron mass polarization}} + V.} \tag{2.11}$$

Why the mass-polarization term?

It is the *price* of referring electrons to the *nuclear* COM (not the total COM): the electron momenta no longer commute additively with M_N. We keep it exactly.

Lifting $\mathbf{P}_{\rm cm} = 0$: a canonical transformation

For $\mathbf{P}_{\rm cm} \neq 0$, (2.10) carries two artifacts:

$$H = \underbrace{\frac{\mathbf{P}_{\rm cm}^2}{2M_N}}_{(A)\text{ wrong mass}} \underbrace{- \frac{\mathbf{P}_{\rm cm} \cdot \sum_i \mathbf{p}'_i}{M_N}}_{(B)\text{ cross-coupling}} + \frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e} + V. \tag{2.12}$$

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A single canonical transformation generated by $G = \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}} \cdot \sum_i \mathbf{r}_i'$ removes both:

$$\tilde{\mathbf{r}}_i' = \mathbf{r}_i', \quad \tilde{\mathbf{p}}_i' = \mathbf{p}_i' - \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}}, \quad \tilde{\mathbf{R}}_{\mathrm{cm}} = \mathbf{R}_{\mathrm{cm}} + \frac{m_e}{M_{\mathrm{tot}}} \sum_i \mathbf{r}_i'. \tag{2.15}$$

Why bother?

It relabels the COM from *nuclear* to *total* (Galilean boost), correcting $M_N \to M_{\mathrm{tot}}$ and killing (B) — *exactly, for any* $\mathbf{P}_{\mathrm{cm}}$. The internal physics is then frame-independent.

The frame-independent internal Hamiltonian

The result is structurally identical to the rest-frame form, but exact for *any* $\mathbf{P}_{\rm cm}$:

$$\boxed{\tilde{H} = \frac{\mathbf{P}_{\rm cm}^{\,2}}{2M_{\rm tot}} + \sum_j \frac{\mathbf{P}_j'^{\,2}}{2M_j} + \sum_i \frac{\tilde{\mathbf{p}}_i'^{\,2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \tilde{\mathbf{p}}_i'\Big)^2 + V(\mathbf{R}_j', \mathbf{r}_i').} \tag{2.22}$$

- COM term now carries the *correct* total mass $M_{\rm tot} = M_N + N_{\rm elec} m_e$.
- Internal part = nuclear KE + electron KE + mass polarization + Coulomb.
- $\mathbf{P}_{\rm cm}$ is a decoupled, conserved spectator.

(2.22) is what we carry into the body-fixed frame.

Body-fixed frame and the Routhian

Body-fixed frame: rotate with the nuclei

Introduce a rotation $\mathbf{S}(\Omega)$ (Euler angles $\Omega = \phi, \theta, \chi$) and body-fixed (BF) coordinates:

$$\mathbf{R}'_j = \mathbf{S}(\Omega)\,\bar{\mathbf{R}}_j(Q), \qquad \mathbf{r}'_i = \mathbf{S}(\Omega)\,\bar{\mathbf{r}}_i. \tag{3.1}$$

The *Eckart* conditions fix $\mathbf{S}$ from the *nuclei*; the nuclear shape is rectilinear mass-weighted normal coordinates Q_b:

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\mathrm{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb}\, Q_b, \qquad \sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}. \tag{3.1a}$$

The BF angular velocity $\boldsymbol{\omega}$ is defined by

$$(\mathbf{S}^T \dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma}\, \omega_\gamma. \tag{3.2}$$

A Routhian: keep electrons in momentum form

We need an action principle for the BF *nuclear* coordinates — but the electronic kinetic energy + mass polarization $H_e = \sum_i \tilde{\mathbf{p}}_i'^2/2m_e + \frac{1}{2M_N}(\sum_i \tilde{\mathbf{p}}_i')^2$ is a *rotational scalar*. So Legendre-transform only the COM and nuclear momenta, leaving the electrons canonical $\Rightarrow$ a Routhian:

$$\boxed{\mathcal{R} = \tfrac{1}{2} M_{\mathrm{tot}} |\dot{\tilde{\mathbf{R}}}_{\mathrm{cm}}|^2 + \tfrac{1}{2} \sum_j M_j |\dot{\mathbf{R}}_j'|^2 - H_e(\tilde{\mathbf{p}}_i') - V\,.} \tag{3.6}$$

Why a Routhian?

It avoids converting the electron kinetic energy to velocity form and back. The electrons ride along as the scalar H_e — a Lagrangian in $(\tilde{\mathbf{R}}_{\mathrm{cm}}, \mathbf{R}_j')$, *minus* a Hamiltonian in $(\mathbf{r}_i', \tilde{\mathbf{p}}_i')$.

Body-fixed Routhian

H_e is built from $\tilde{\mathbf{p}}'_i \cdot \tilde{\mathbf{p}}'_l$; under the passive rotation $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T \tilde{\mathbf{p}}'_i$ it is *unchanged* in form.
Substituting the BF velocities $\dot{\mathbf{R}}'_j = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j)$,

$$\mathcal{R} = \tfrac{1}{2} M_{\mathrm{tot}} |\dot{\tilde{\mathbf{R}}}_{\mathrm{cm}}|^2 + \tfrac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N} \Big(\sum_i \bar{\mathbf{p}}_i \Big)^2 - V. \tag{3.7}$$

Only the *nuclear* kinetic term carries $\boldsymbol{\omega}$; V is now manifestly **S**-independent.

Expand the nuclear term

The vector identity (for any $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega}$)

$$\frac{m}{2}|\boldsymbol{\omega}\times\mathbf{a}+\dot{\mathbf{a}}|^2 = \underbrace{\frac{m}{2}\boldsymbol{\omega}^T(a^2\mathbb{1}-\mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m\,\boldsymbol{\omega}\cdot(\mathbf{a}\times\dot{\mathbf{a}})}_{\text{Coriolis}} + \underbrace{\frac{m}{2}|\dot{\mathbf{a}}|^2}_{\text{vibrational}} \tag{3.9c}$$

Expand the nuclear term

The vector identity (for any $\mathbf{a}, \dot{\mathbf{a}}, \omega$)

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applied to the nuclei ($m\to M_j$, $\mathbf{a}\to\bar{\mathbf{R}}_j$) gives the assembled Routhian

$$\boxed{\mathcal{R} = \tfrac{1}{2}M_{\mathrm{tot}}|\dot{\tilde{\mathbf{R}}}_{\mathrm{cm}}|^2 + \underbrace{\tfrac{1}{2}\omega^T\mathbf{I}_N\omega + \omega\cdot\mathbf{L}_N^{\mathrm{vib}} + T_{\mathrm{vib}}^N}_{\text{nuclear}} - H_e - V.} \tag{3.13}$$

The rotational kernel is the *nuclear* inertia $\mathbf{I}_N$ alone — no electronic inertia, because the electrons were never expanded in ω.

Canonical momenta and the total angular momentum

Nuclear angular momentum (from $\partial\mathcal{R}/\partial\boldsymbol{\omega}$):

$$\mathbf{J}_N = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}}. \tag{3.15}$$

The BF frame rotates nuclei *and* electrons, so the total angular momentum is the sum [$\mathbf{L}_{\mathrm{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$]:

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\mathrm{elec}} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}} + \mathbf{L}_{\mathrm{elec}}. \tag{3.16}$$

Vibrational momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b$:

$$P_b = \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b + \dot{Q}_b, \qquad \boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a, \quad \boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}. \tag{3.17a,b}$$

The key identity

Subtract the electronic part from (3.16):

$$\boxed{\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q)\,\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}.} \tag{3.19}$$

Why this matters

The right side is *purely nuclear*: the rotational kernel is $\mathbf{I}_N$, **not** the full inertia. All electronic + mass-polarization content sits on the left, in the kinematic shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\text{elec}}$.

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Cross-check: the velocity-form route instead generates the electronic inertia $\mathbf{I}_e, \mathbf{I}_X$ in *both* $\mathbf{J}$ and $\mathbf{L}_{\mathrm{elec}}$ — they cancel exactly, reproducing (3.19). The Routhian makes the cancellation automatic.

Watson reciprocal inertia

Invert (3.19) for $\boldsymbol{\omega}$: write $\mathbf{L}_N^{\mathrm{vib}} = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\omega}$ with

$$\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b \;\; \text{(field-free vibrational ang. mom.)}, \qquad \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}} = \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^{\mathsf{T}}. \tag{3.20a}$$

Then $\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})\boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boxed{\boldsymbol{\omega} = \boldsymbol{\mu}\big(\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi}\big), \qquad \boldsymbol{\mu} \equiv (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}.} \tag{3.20}$$

$\boldsymbol{\mu}$ is the Watson reciprocal-inertia tensor — built from *nuclear* quantities only, with no Wilson G-matrix.

The exact classical molecular Hamiltonian

Legendre back; the rovibrational kinetic energy becomes $\frac{1}{2}(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi})^{\mathsf{T}}\boldsymbol{\mu}(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi})+\frac{1}{2}\sum_b P_b^2$:

$$\boxed{\begin{aligned} H_{\mathrm{int}} = {} & \tfrac{1}{2}\left(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi}\right)^{\mathsf{T}}\boldsymbol{\mu}(Q)\left(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi}\right)+\tfrac{1}{2}\sum_b P_b^2 \\ & +\sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e}+\frac{1}{2M_N}\left(\sum_i \bar{\mathbf{p}}_i\right)^2+V_{NN}+V_{Ne}+V_{ee}. \end{aligned}} \tag{3.22}$$

Exact, classical, in BF + Eckart coordinates. Now we quantize.

The quantization problem

Why naive quantization fails in curvilinear coordinates

Label the configuration coordinates and write the classical kinetic energy as a quadratic form:

$$\{q^A\} = \{\phi, \theta, \chi,\ Q_1, \ldots, Q_{3N_{\mathrm{nucl}}-6},\ \bar{\mathbf{r}}_1, \ldots\}, \qquad T = \tfrac{1}{2}\, g^{AB}(q)\, \pi_A \pi_B. \tag{4.1, 4.2}$$

The metric g_{AB} inherits the block structure of (3.22): Watson $\boldsymbol{\mu}(Q)$ (rotation), identity (vibration), Cartesian (electrons).

The problem

The Euler angles and Q are *curvilinear*. The natural Hilbert space is $L^2\big(\sqrt{|g|}\, d^n q\big)$, but $\hat{\pi}_A = -i\hbar\partial_A$ makes $\frac{1}{2} g^{AB}\hat{\pi}_A\hat{\pi}_B$ **non-Hermitian** there (g^{AB} depends on q and does not commute with ∂_A).

Podolsky quantization

The Podolsky / Laplace–Beltrami prescription

The unique kinetic operator Hermitian on $L^2(\mathcal{C}, \sqrt{|g|}\, d^n q)$ is the Laplace–Beltrami operator:

$$\boxed{\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{|g|}} \partial_A\big(\sqrt{|g|}\, g^{AB}\, \partial_B\big).} \tag{4.3}$$

Equivalently, with the *Hermitian* momenta $\hat{\pi}_A = -i\hbar\big(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|}\big)$, one has $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB} \hat{\pi}_B$.

Why this form?

The $\sqrt{|g|}$ factors symmetrize the operator about the curved measure. The price: a residual Q-dependent c-number — the Watson pseudopotential — once we flatten the measure.

The molecular volume element

Building g_{AB} from the velocity-form kinetic energy and reducing the determinant (Schur complements over the electron and vibrational blocks) gives

$$\boxed{\sqrt{|g|}\, d^n q = \underbrace{\sin\theta\, d\phi\, d\theta\, d\chi}_{\text{Haar on } SO(3)} \cdot \underbrace{\sqrt{\det \mathbf{I}'(Q)}\, \prod_b dQ_b}_{\text{rot.–vib. Jacobian}} \cdot \underbrace{\prod_i d^3\bar{\mathbf{r}}_i}_{\text{flat}}\, .} \tag{4.4}$$

The Jacobian carries the Watson modified inertia $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^\top$, *not* the bare $\mathbf{I}_N$: the rotation–vibration Coriolis coupling reduces $\det \mathbf{I}_N \to \det \mathbf{I}'$.

Rescaling to a flat measure

To work on the simpler measure $d\mu_W = \sin\theta \, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$, rescale the wavefunction by the Q-Jacobian $\gamma \equiv \det \mathbf{I}'$:

$$\psi_W \equiv \left(\det \mathbf{I}'(Q)\right)^{1/4}\psi, \qquad \hat{T}_W = \mathcal{J}^{1/2}\,\hat{T}\,\mathcal{J}^{-1/2}, \quad \mathcal{J} = (\det \mathbf{I}')^{1/2}. \tag{4.5}$$

Why rescale?

$\psi_W \in L^2(d\mu_W)$ with the *flat* vibrational measure, so $\hat{P}_b = -i\hbar\partial_{Q_b}$ is Hermitian. The similarity does not commute through $\gamma(Q)$ — the leftover is exactly the pseudopotential $\hat{U}(Q)$.

Anomalous body-fixed angular momentum

Body-fixed angular momentum has a sign flip

Total **J** is an SF observable; its BF components are projections through $\mathbf{S}(\Omega)$:

$$\hat{J}^{\mathrm{BF}}_{\alpha} = \mathbf{S}_{\beta\alpha}\,\hat{J}^{\mathrm{SF}}_{\beta}. \tag{4.6}$$

But $\mathbf{S}(\Omega)$ depends on the Euler angles that $\hat{J}^{\mathrm{SF}}$ differentiates. Commuting $\hat{J}^{\mathrm{SF}}$ past **S** adds terms that *outweigh and flip* the usual algebra:

$$\boxed{[\hat{J}^{\mathrm{BF}}_{\alpha}, \hat{J}^{\mathrm{BF}}_{\beta}] = -\,i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{J}^{\mathrm{BF}}_{\gamma}.} \tag{4.7}$$

The famous anomalous commutator (Klein; Van Vleck): note the − sign.

Normal vs. anomalous; commuting sectors

The electronic angular momentum is built intrinsically from BF electron operators — *normal* algebra:

$$[\hat{L}_{\mathrm{elec},\alpha}, \hat{L}_{\mathrm{elec},\beta}] = +i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{L}_{\mathrm{elec},\gamma}, \qquad [\hat{J}^{\mathrm{BF}}_{\alpha}, \hat{L}_{\mathrm{elec},\beta}] = 0. \tag{4.8, 4.9}$$

Why it (barely) matters

The sign flip controls the ordering of $\hat{J}_\alpha \hat{J}_\beta$ in the kinetic operator. But contracted with the *symmetric* $\mu_{\alpha\beta}$ the antisymmetric commutator drops out ($\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$) — so it contributes *no* extra potential. We use this next.

Watson pseudopotential

The general curvilinear extra-potential

Conjugating the Laplace–Beltrami operator $\hat{T}_{\mathrm{LB}} = -\frac{\hbar^2}{2} g^{-1/2}\partial_A(g^{1/2}g^{AB}\partial_B)$ by $g^{1/4}$ and collecting terms (first-order pieces cancel by symmetry of g^{AB}) gives the exact, coordinate-general result

$$\boxed{\hat{T}_W = \tfrac{1}{2}\,\hat{p}_A\, g^{AB}\,\hat{p}_B + U, \qquad U = \frac{\hbar^2}{8}\left[\tfrac{1}{4}g^{AB}(\partial_A \ln g)(\partial_B \ln g) + \partial_A(g^{AB}\partial_B \ln g)\right].} \tag{4.10d}$$

Why $\hbar^2/8$?

The two half-powers $g^{\pm 1/4}$ of the rescaling each contribute $\frac{1}{2}\partial \ln g$; their product/derivative is $\mathcal{O}(\hbar^2)$ with prefactor $\frac{1}{8}$.

The molecular metric and its determinant

In the body-frame momentum basis $(\hat{J}_\alpha, \hat{P}_b)$ [with $\boldsymbol{\pi} = \boldsymbol{\sigma}^\top \hat{P}$], read off

$$g^{AB} = \begin{pmatrix} \boldsymbol{\mu} & -\boldsymbol{\mu}\boldsymbol{\sigma}^\top \\ -\boldsymbol{\sigma}\boldsymbol{\mu} & \mathbb{1} + \boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^\top \end{pmatrix}, \qquad \det g^{AB} = \det \boldsymbol{\mu} \tag{4.10e,f}$$

(Schur complement of the $\boldsymbol{\mu}$ block). Hence $g = 1/\det \boldsymbol{\mu} = \det \mathbf{I}' \equiv \gamma(Q)$. Since the $\hat{J}_\alpha$ already carry $\sin\theta$, only $\gamma(Q)$ is rescaled: set $\ln g \to \ln \gamma$ in (4.10d).

Rotational sector: no c-number

The only ordering ambiguity is in $\hat{J}_\alpha \hat{J}_\beta$. Split and use the anomalous algebra (4.7):

$$\tfrac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta = \tfrac{1}{4}\mu_{\alpha\beta}\{\hat{J}_\alpha, \hat{J}_\beta\} - \frac{i\hbar}{4}\,\mu_{\alpha\beta}\,\epsilon_{\alpha\beta\gamma}\,\hat{J}_\gamma. \tag{4.10g}$$

Symmetric $\mu_{\alpha\beta}$ × antisymmetric $\epsilon_{\alpha\beta\gamma} = 0 \Rightarrow$ the commutator term vanishes. The rotational kernel is Hermitian as written, with no extra potential.

Vibrational sector: the rescaling, step by step

With $g^{ab} = \delta_{ab}$ but measure $\gamma(Q)$, the Podolsky vibrational operator is $\hat{T}_{\rm vib} = -\frac{\hbar^2}{2}\gamma^{-1/2}\sum_b \partial_{Q_b}(\gamma^{1/2}\partial_{Q_b})$. Conjugate by $\gamma^{1/4}$, working outward (one product rule per line):

$$\begin{aligned}
\partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{-1/4}\left[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W\right],\\
\gamma^{1/2}\partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{1/4}\left[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W\right],\\
\partial_{Q_b}\big(\gamma^{1/4}[\cdots]\big) &= \gamma^{1/4}\left[\partial^2_{Q_b}\psi_W - \tfrac{1}{16}(\partial_{Q_b}\ln\gamma)^2\psi_W - \tfrac{1}{4}(\partial^2_{Q_b}\ln\gamma)\psi_W\right].
\end{aligned}$$

The cross terms $\pm\frac{1}{4}(\partial\ln\gamma)\partial\psi_W$ cancel. Result:

$$\gamma^{1/4}\,\hat{T}_{\rm vib}\gamma^{-1/4} = \frac{1}{2}\sum_b \hat{P}_b^2 + U_{\rm vib}, \quad U_{\rm vib} = \frac{\hbar^2}{8}\sum_b\left[\partial^2_{Q_b}\ln\gamma + \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2\right]. \tag{4.10k}$$

Collapse to the Watson trace

Two exact identities finish the job. **Jacobi:**

$$\partial_b \ln \gamma = \mathrm{tr}(\boldsymbol{\mu}\,\partial_b \mathbf{I}'), \qquad \partial_b \boldsymbol{\mu} = -\boldsymbol{\mu}(\partial_b \mathbf{I}')\boldsymbol{\mu}. \tag{4.10l}$$

Linearity: $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$ with $\boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a$, so $\partial_b \mathbf{I}'$ is built from *constant* ζ's. Substituting, the gradient terms cancel and the Coriolis sum rules collapse everything to a single trace:

$$\boxed{\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^{3} \mu_{\alpha\alpha}(Q).} \tag{4.11}$$

A pure c-number from the *measure* — the electrons do not modify it.

The complete quantum Hamiltonian

Electronic kinetic operator

In BF Cartesian electron coordinates $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is already Hermitian on the flat measure, so the electron + mass-polarization piece quantizes directly:

$$\hat{T}_e = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2\mu_e} + \frac{1}{M_N}\sum_{i<l} \hat{\bar{\mathbf{p}}}_i \cdot \hat{\bar{\mathbf{p}}}_l. \qquad (4.12,\ 4.13)$$

The diagonal part renormalizes the electron mass, $\frac{1}{m_e} + \frac{1}{M_N} = \frac{1}{\mu_e}$; the off-diagonal two-electron term is the mass polarization *sensu stricto* (nuclear recoil).

The exact non-relativistic molecular Hamiltonian

Collecting the rovibrational kernel (3.22), the Watson pseudopotential (4.11), and $\hat{T}_e$:

$$\boxed{\begin{aligned}
\hat{H}_{\rm int} = {} & \tfrac{1}{2}\sum_{\alpha\beta}\big(\hat{J}_\alpha - \hat{L}_{{\rm elec},\alpha} - \hat{\pi}_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat{J}_\beta - \hat{L}_{{\rm elec},\beta} - \hat{\pi}_\beta\big) && \text{rotation–vibration–electron} \\
& + \tfrac{1}{2}\sum_b \hat{P}_b^2 && \text{vibration} \\
& + \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^{\,2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 && \text{electrons + mass pol.} \\
& - \frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}(Q) && \text{Watson pseudopotential} \\
& + V_{NN} + V_{Ne} + V_{ee} && \text{Coulomb}
\end{aligned}} \qquad (4.14)$$

Hermitian on $L^2(\mathcal{C}, d\mu_W)$, $\; d\mu_W = \sin\theta\, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$.

Summary

The journey, in four stages

1. **Classical**: lab H (1.1) $\to$ frame-independent internal $\tilde{H}$ (2.22), COM split off exactly for any $\mathbf{P}_{\rm cm}$.
2. **Body-Fixed**: Routhian (electrons stay canonical) $\to$ key identity $\mathbf{J} - \mathbf{L}_{\rm elec} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\rm vib} \to$ classical $H_{\rm int}$ (3.22) with Watson $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^\top)^{-1}$.
3. **Quantize**: Podolsky operator (4.3) + rescaling (4.5); anomalous algebra $[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$; Watson pseudopotential $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$.
4. **Quantum $\hat{H}$**: the exact non-relativistic $\hat{H}_{\rm int}$ (4.14).

What we did — and did not — assume

$\hat{H}_{\mathrm{int}}$ (4.14) is exact. The only inputs were:

- non-relativistic dynamics;
- the Eckart frame (a convention);
- $3N_{\mathrm{nucl}} - 6$ internal coordinates (an exact parametrization).

No Born–Oppenheimer, no clamped nuclei, no neglect of mass polarization, no harmonic assumption.

Outlook

(4.14) is the rigorous starting point: a Born–Huang expansion in the parametric electronic basis, then the Born–Oppenheimer truncation, recovers the familiar single-surface picture — a controlled approximation *of an exact operator*.

The full molecular Hamiltonian:

exact, quantized, and ready to approximate.